# P-BRIDGE

## eSim One-Click Simulation Bridge for KiCad 8.0

*Complete User Guide -Installation, Setup & Usage*

---

**Developed by: Imran Farhat**

B.Tech CSE (AI/ML) | FOSSEE Semester Long Internship Spring 2026 | IIT Bombay

GitHub: https://github.com/ImranFarhat01/esim-bridge-plugin

Version 1.0.0 | March 2026

# Chapter 1 -What is P-BRIDGE?

P-BRIDGE is a KiCad 8.0 plugin that connects KiCad schematic design directly to eSim 2.5 circuit simulation. It automates the entire workflow from schematic to waveform graph with a single button click.

## 1.1 The Problem it Solves

Without P-BRIDGE, simulating a KiCad circuit in eSim requires 26 manual steps taking 10–15 minutes. Every time you change a component value, you repeat the entire process.

## 1.2 What P-BRIDGE Does

- Exports the KiCad netlist automatically using kicad-cli
- Converts it to a valid SPICE file with correct net names
- Writes all required eSim project files automatically
- Launches eSim with the project ready to simulate
- Reduces the 10 - 15 step process to 4 steps in 30 seconds

## 1.3 Supported Analysis Types

| Analysis | SPICE Command & Description |
|---|---|
| **Transient (.tran)** | Voltage/current vs time -most common |
| **AC Sweep (.ac)** | Gain/phase vs frequency |
| **DC Sweep (.dc)** | Output vs DC source voltage |
| **Operating Point (.op)** | Single DC bias point -results shown in popup |

## 1.4 Supported Components

| Component | SPICE Format |
|---|---|
| **Resistor (R)** | R1 node1 node2 10k |
| **Capacitor (C)** | C1 node1 node2 100nF |
| **Inductor (L)** | L1 node1 node2 1mH |
| **DC Voltage Source (V)** | V1 n+ n- DC 5 |
| **Sine Source (VSIN)** | V1 n+ n- AC 1 SIN(0 1 1k) |
| **Pulse Source (PULSE)** | V1 n+ n- PULSE(0 5 0 1n 1n 5m 10m) |
| **Current Source (I)** | I1 n+ n- DC 1m |
| **Diode / LED (D)** | D1 anode cathode dled [generic model auto-injected] |
| **BJT Transistor (Q)** | Q1 collector base emitter model |

| Component | SPICE Format |
|---|---|
| **MOSFET (M)** | M1 drain gate source bulk model |
| **Digital IC (U/X)** | Commented out -needs external .subckt model |

---

## Chapter 2 -System Requirements

⚠ **WSL WARNING:** WSL (Windows Subsystem for Linux) has limited GUI support. The plugin requires KiCad GUI, eSim GUI, and ngspice -all graphical applications. WSL with WSLg (Windows 11 or recent Windows 10) may work but is not guaranteed. VirtualBox Ubuntu is the fully tested and recommended environment.

### 2.1 Environment Options

| Environment | Status |
|---|---|
| **Ubuntu 24.04 in VirtualBox (Windows host)** | ✅ Fully tested -recommended |
| **Native Ubuntu 24.04 Linux** | ✅ Works perfectly |
| **WSL 2 with WSLg (Windows 11)** | ⚠ May work -follow WSL-specific steps |
| **WSL 1 or WSL without WSLg** | ❌ Will not work -no display support |
| **macOS** | ❌ Not supported -eSim 2.5 is Linux only |

### 2.2 Software Requirements

| Software | Version / Notes |
|---|---|
| **Ubuntu Linux** | 24.04 LTS |
| **KiCad** | 8.0 (must include kicad-cli) |
| **eSim** | 2.5 installed at ~/Downloads/eSim-2.5/ |
| **ngspice** | 35+ (bundled with eSim, also needs system package) |
| **Python** | 3.10+ (comes with Ubuntu) |
| **git** | Any version |

### 2.3 Hardware Requirements (VirtualBox)

| Setting | Minimum / Recommended |
|---|---|
| **RAM (Windows host)** | 8 GB minimum, 16 GB recommended |
| **RAM allocated to VM** | 4 GB minimum, 8 GB recommended |
| **Disk space for VM** | 40 GB |
| **Graphics Controller** | VMSVGA (critical -not VBoxVGA) |
| **Video Memory** | 256 MB |
| **3D Acceleration** | Enabled |

---

## Chapter 3 -Installation on VirtualBox Ubuntu (Recommended)

✅ **Use this chapter if:** You are on Windows and want to use VirtualBox -the fully tested, guaranteed-to-work environment.

| STEP 1 | Set Up VirtualBox with Ubuntu 24.04 |
|---|---|

### Download and install VirtualBox

1. Download VirtualBox from https://virtualbox.org and install it on Windows
2. Download Ubuntu 24.04 LTS ISO from https://ubuntu.com/download/desktop
3. Create a new VM in VirtualBox with these settings:

| Setting | Value |
|---|---|
| **Name** | Ubuntu 24.04 eSim |
| **Type** | Linux -Ubuntu (64-bit) |
| **RAM** | 4096 MB minimum (8192 recommended) |
| **CPU** | 2 cores or more |
| **Storage** | 40 GB VDI dynamically allocated |
| **Graphics Controller** | VMSVGA ← critical, not VBoxVGA |
| **Video Memory** | 256 MB |
| **3D Acceleration** | Enabled |

4. Start the VM, point it to the Ubuntu ISO, install Ubuntu normally
5. After Ubuntu installs, log in and open a terminal (Ctrl+Alt+T)

### Install VirtualBox Guest Additions (for better display)

```
sudo apt update && sudo apt install -y build-essential dkms linux-headers-$(uname -r)
# In VirtualBox menu: Devices → Insert Guest Additions CD Image
sudo /media/$USER/VBox_GAs_*/VBoxLinuxAdditions.run
sudo reboot
```

## STEP 2 Install KiCad 8.0

```
sudo apt update
sudo apt install -y kicad
kicad-cli --version
# Expected: Application: kicad-cli 8.0.x
```

## STEP 3 Install eSim 2.5

```
cd ~/Downloads
wget https://static.fossee.in/esim/installation-files/eSim-2.5.zip
unzip eSim-2.5.zip
cd eSim-2.5
chmod +x install-eSim.sh
./install-eSim.sh --install
# Wait 5-10 minutes for installation to complete
```

Verify eSim installed correctly:

```
ls ~/Downloads/eSim-2.5/src/frontEnd/Application.py
ls ~/.esim/env/bin/python3
# Both must print the file path -if not, reinstall eSim
```

### Test eSim launches correctly

```
cd ~/Downloads/eSim-2.5/src/frontEnd
PYTHONPATH=/home/$(whoami)/Downloads/eSim-2.5/src ~/.esim/env/bin/python3
Application.py
```

eSim should open with toolbar icons visible. Close it after confirming.

## STEP 4 Install the P-BRIDGE Plugin

```
cd ~
git clone https://github.com/ImranFarhat01/esim-bridge-plugin.git
cp -r ~/esim-bridge-plugin/esim_bridge ~/.local/share/kicad/8.0/scripting/plugins/
ls -la ~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/
# Must show esim_bridge.py (~50KB), icon.png, __init__.py
```

## STEP 5 Fix Username in Plugin

⚠ **This step is mandatory.** The plugin has hardcoded paths with the developer's username 'imran-farhat'. You must replace it with your own username or the plugin will fail to find eSim.

```
# Replace developer username with your username
sed -i "s/imran-farhat/$(whoami)/g"
~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/esim_bridge.py
# Verify -must return NO output (no lines containing old username)
grep "imran-farhat" ~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/esim_bridge.py
```

## STEP 6 Fix __init__.py and Create eSim Workspace

```
# Fix __init__.py
echo "from .esim_bridge import ESimBridgePlugin" >
~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/__init__.py
cat ~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/__init__.py
# Must show: from .esim_bridge import ESimBridgePlugin
```

```
# Create eSim workspace
mkdir -p ~/eSim-Workspace
echo '{"'/home/$(whoami)/eSim-Workspace/esim_bridge_project'": []}' >
~/eSim-Workspace/.projectExplorer.txt
```

| STEP 7 | Launch KiCad and Verify Plugin |
|---|---|

6. Open KiCad from Applications menu
7. Open PCB Editor (File → New → PCB or click the green PCB icon)
8. Look for the eSim Bridge icon in the PCB Editor toolbar
9. If not visible: Tools → External Plugins → Refresh Plugins

✅ **Plugin loaded successfully if:** You see the eSim FOSSEE logo icon in the toolbar, or 'eSim Simulation Bridge' appears under Tools → External Plugins.

---

## Chapter 4 -Installation on WSL Ubuntu (Windows Subsystem for Linux)

ℹ **Use this chapter if:** You want to use WSL instead of VirtualBox. Requires Windows 11 or Windows 10 with WSLg support. GUI apps must be able to open windows from WSL.

| STEP 1 | Install KiCad 8.0 (with ngspice conflict fix) |
|---|---|

KiCad and system ngspice conflict on Ubuntu 24.04. Follow these steps in exact order:

```
sudo apt update
sudo apt install kicad -y
# If error about libngspice-kicad conflict -run these:
sudo dpkg --remove --force-depends ngspice
sudo dpkg -i --force-overwrite /var/cache/apt/archives/libngspice-kicad_*.deb
sudo apt-get install -f -y
# Now install ngspice back
sudo apt install ngspice -y
sudo dpkg -i --force-overwrite /var/cache/apt/archives/ngspice_*.deb
```

```
sudo apt-get install -f -y
# Verify both work
kicad-cli --version
ngspice --version
```

## STEP 2 Verify KiCad GUI Opens in WSL

```
kicad
# KiCad window must appear -if you see display errors, WSLg is not working
# Close KiCad after confirming it opens
```

⚠ **If KiCad shows display errors:** Your WSL does not support GUI apps. You must use VirtualBox instead (Chapter 3). There is no workaround for WSL without display support.

## STEP 3 Install eSim 2.5

```
cd ~/Downloads
wget https://static.fossee.in/esim/installation-files/eSim-2.5.zip
unzip eSim-2.5.zip
cd eSim-2.5
chmod +x install-eSim.sh
./install-eSim.sh --install
# Wait 5-10 minutes
```

Verify:

```
ls ~/Downloads/eSim-2.5/src/frontEnd/Application.py
ls ~/.esim/env/bin/python3
```

## STEP 4 Install Plugin, Fix Username, Setup Workspace

These steps are identical to the VirtualBox installation:

```
# Install plugin
cd ~
git clone https://github.com/ImranFarhat01/esim-bridge-plugin.git
cp -r ~/esim-bridge-plugin/esim_bridge ~/.local/share/kicad/8.0/scripting/plugins/

# Fix username
sed -i "s/imran-farhat/$(whoami)/g"
~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/esim_bridge.py
grep "imran-farhat" ~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/esim_bridge.py
# Must return nothing

# Fix __init__.py
echo "from .esim_bridge import ESimBridgePlugin" >
~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/__init__.py

# Create workspace
mkdir -p ~/eSim-Workspace
echo '{"'/home/$(whoami)/eSim-Workspace/esim_bridge_project'": []}' >
~/eSim-Workspace/.projectExplorer.txt

# Launch KiCad and verify plugin appears
kicad
```

---

## Chapter 5 -Using the Plugin

### 5.1 Create a Test Circuit in KiCad

Before simulating your real circuit, test with a simple voltage divider -the proven working test circuit.

#### Open Schematic Editor

10. In KiCad: File → New → Project → choose a folder
11. Double-click the .kicad_sch file to open Schematic Editor

#### Place Components (press A to open component search)

- Search **R** → place R1 and R2
- Double-click R1 → set Value: **10k**
- Double-click R2 → set Value: **10k**
- Search **VSIN** → place V1
- Search **GND** → place 3 ground symbols

### Configure Voltage Source V1 (critical)

Double-click V1 → find Simulation Parameters and set:

```
Sim.Type:   SIN
Sim.Params: dc=0 ampl=1 f=1k ac=1
```

### Connect the Circuit

12. Press W to draw wires: V1 positive → R1 top → R1 bottom → R2 top → R2 bottom → GND
13. Press L → type vmid → place net label at the junction between R1 and R2
14. Press Ctrl+S to save

## 5.2 Run the Simulation

> ℹ **Important:** The plugin button appears in the PCB Editor toolbar, not the Schematic Editor. Switch to PCB Editor before clicking the plugin.

15. Switch to KiCad PCB Editor
16. Click the eSim Bridge icon in the toolbar
17. The Analysis Configuration dialog appears -select analysis type and parameters
18. Click Simulate in eSim →
19. Review the SPICE preview → click Launch eSim →
20. In eSim: double-click esim_bridge_project in the left panel
21. Click Simulate
22. Click Plot → select a node → graph appears

## 5.3 Analysis Parameters Reference

### Transient Analysis

| Parameter | Default / Description |
|---|---|
| **Start Time** | 0 -simulation start |
| **Step Time** | 1us -time resolution (use units: us, ms, ns, s) |
| **Stop Time** | 10ms -simulation end time |

### AC Analysis

| Parameter | Default / Description |
|---|---|
| **Scale** | dec -logarithmic decades (also: lin, oct) |
| **Start Frequency** | 1Hz |

| Parameter | Default / Description |
|---|---|
| **Stop Frequency** | 1MEGHz |
| **No. of Points** | 100 per decade |

### DC Sweep

| Parameter | Default / Description |
|---|---|
| **Source Name** | V1 -name of DC source to sweep |
| **Start Value** | 0 V |
| **Stop Value** | 5 V |
| **Step** | 0.1 V increment |

### Operating Point

No parameters needed. Plugin runs ngspice directly and shows DC node voltages in a popup. No waveform graph is produced.

---

## Chapter 6 -Complete Command Reference

### 6.1 One-Shot Install Script (copy and run all at once)

For a clean Ubuntu system with no prior KiCad or eSim installation:

```
# === P-BRIDGE COMPLETE INSTALLATION SCRIPT ===

# Step 1: Install KiCad
sudo apt update && sudo apt install -y kicad git

# Step 2: Install eSim 2.5
cd ~/Downloads
wget https://static.fossee.in/esim/installation-files/eSim-2.5.zip
unzip eSim-2.5.zip && cd eSim-2.5
chmod +x install-eSim.sh && ./install-eSim.sh --install

# Step 3: Install plugin
cd ~ && git clone https://github.com/ImranFarhat01/esim-bridge-plugin.git
cp -r ~/esim-bridge-plugin/esim_bridge ~/.local/share/kicad/8.0/scripting/plugins/

# Step 4: Fix username
```

```
sed -i "s/imran-farhat/$(whoami)/g"
~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/esim_bridge.py

# Step 5: Fix __init__.py
echo "from .esim_bridge import ESimBridgePlugin" >
~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/__init__.py

# Step 6: Create workspace
mkdir -p ~/eSim-Workspace
echo '{"'/home/$(whoami)/eSim-Workspace/esim_bridge_project'": []}' >
~/eSim-Workspace/.projectExplorer.txt

echo 'Installation complete! Launch KiCad with: kicad'
```

### 6.2 Verification Commands

```
# Check all components are installed
kicad-cli --version
ngspice --version
ls ~/.esim/env/bin/python3
ls ~/Downloads/eSim-2.5/src/frontEnd/Application.py
ls ~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/
cat ~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/__init__.py
```

### 6.3 Plugin Management

```
# Reinstall plugin after update
cp -r ~/esim-bridge-plugin/esim_bridge ~/.local/share/kicad/8.0/scripting/plugins/

# View plugin log
cat ~/.local/share/kicad/esim_bridge.log

# Follow log in real time
tail -f ~/.local/share/kicad/esim_bridge.log

# Fix empty __init__.py
echo "from .esim_bridge import ESimBridgePlugin" >
~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/__init__.py
```

### 6.4 Simulation Files

```
# View generated SPICE file
cat ~/eSim-Workspace/esim_bridge_project/esim_bridge_project.cir.out

# Check analysis type written by plugin
cat ~/eSim-Workspace/esim_bridge_project/analysis

# Check simulation voltage output
head -20 ~/eSim-Workspace/esim_bridge_project/plot_data_v.txt

# Run ngspice manually for debugging
ngspice -b ~/eSim-Workspace/esim_bridge_project/esim_bridge_project.cir.out

# Delete stale .raw file manually
rm -f ~/eSim-Workspace/esim_bridge_project/esim_bridge_project.raw
```

### 6.5 Launch eSim Manually

```
cd ~/Downloads/eSim-2.5/src/frontEnd
PYTHONPATH=/home/$(whoami)/Downloads/eSim-2.5/src ~/.esim/env/bin/python3
Application.py
```

---

## Chapter 7 -Troubleshooting

| Problem | Cause | Fix Command / Action |
|---|---|---|
| **Plugin not in toolbar** | KiCad not restarted | Restart KiCad → Tools → External Plugins → Refresh Plugins |
| **esim_bridge.py is 0 bytes** | Git clone got empty file | rm -rf ~/esim-bridge-plugin && git clone https://github.com/ImranFarhat01/esim-bridge-plugin.git |
| **__init__.py is empty** | File got corrupted | echo "from .esim_bridge import ESimBridgePlugin" > ~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/__init__.py |
| **eSim not found warning** | eSim not at expected path | ls ~/Downloads/eSim-2.5/src/frontEnd/Application.py |
| **kicad-cli not found** | KiCad not installed | sudo apt install kicad -y |

| Problem | Cause | Fix Command / Action |
|---|---|---|
| **ngspice not found** | ngspice removed during KiCad fix | sudo dpkg -i --force-overwrite /var/cache/apt/archives/ngspice_*.deb && sudo apt-get install -f -y |
| **libngspice-kicad conflict** | Both KiCad and system ngspice own same file | sudo dpkg --remove --force-depends ngspice && sudo dpkg -i --force-overwrite /var/cache/apt/archives/libngspice-kicad_*.deb |
| **Paths still say imran-farhat** | sed command not run | sed -i "s/imran-farhat/$(whoami)/g" ~/.local/share/kicad/8.0/scripting/plugins/esim_bridge/esim_bridge.py |
| **No schematic found** | No project open in KiCad | Open a KiCad project before clicking the plugin button |
| **eSim opens with blank icons** | Wrong VirtualBox display | Set VM display to VMSVGA + 256 MB video memory in VirtualBox settings |
| **Please select project first** | eSim loaded but no project selected | Double-click 'esim_bridge_project' in eSim left panel |
| **UTF-8 popup after simulate** | Stale .raw binary file | Dismiss popup and click Simulate again -benign issue |
| **Flat graph at 0V** | No voltage source in circuit | Add VSIN source with Sim.Type=SIN and Sim.Params set |
| **KiCad display error in WSL** | WSLg not supported | Use VirtualBox instead (Chapter 3) |

---

## Chapter 8 -Known Limitations

| Limitation | Details |
|---|---|
| **OP analysis -no graph** | eSim 2.5 cannot plot .op results. Node voltages shown in popup instead. |
| **UTF-8 popup (cosmetic)** | ngspice shows error when reading stale .raw binary file. Dismiss and re-simulate. |
| **Manual project selection** | eSim requires double-clicking the project in its left panel after opening. |

| Limitation | Details |
|---|---|
| **Linux only** | Plugin uses Linux paths. Not compatible with Windows or macOS. |
| **Single project folder** | All schematics share one eSim project folder (esim_bridge_project). |
| **Basic components only** | Digital ICs (7400, 7402) and components needing .subckt model files are commented out. |
| **Username hardcoded** | Developer username must be replaced using the sed command in Step 5. |

---

## Chapter 9 -Quick Reference Card

**Cut out and keep this card for quick reference:**

**P-BRIDGE -4-Step Simulation Workflow**

1. Open KiCad → draw schematic → save
2. Switch to PCB Editor → click eSim Bridge icon
3. Select analysis type → click 'Simulate in eSim →'
4. In eSim: double-click project → Simulate → Plot

GitHub: github.com/ImranFarhat01/esim-bridge-plugin

---